Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/31/2005
PI Name:	Strayer, Richard Ph.D.		
Project Title:	Denitrification Composter to Stabilize Space-Mission Trash		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	321-861-2928
Organization Name:	Dynamac Corporation		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	03/01/2003	**End Date:**	09/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Garland, Jay (Dynamac Corporation) Reid-Black, Kristina (Dynamac Corporation) Rector, Tony (Dynamac Corporation)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Treatment of space solid wastes, or trash, is a critical issue for mid to long duration space habitation or transportation. Solid wastes are not stable because they are wet (40 – 90 % moisture) and contain levels of soluble organic compounds that can contribute to the growth of undesirable microorganisms with concomitant production of noxious gases and foul-smelling odors. The purpose of the proposed research project is to determine the feasibility of using a denitrifying composter to stabilize trash from space-habitation (STS, ISS, ALS) life support activities. The method of approach starts with design of the denitrification composter, with design criteria derived from experimental variables we wish to manipulate and variables to be held constant. Key environmental and bioprocess parameters for evaluation are nitrate concentration and rate of addition, oxygen concentration, mixing duration and frequency, and inoculum. Parameters to be held constant are pH, temperature, moisture, C:N, solid waste feed material, size reduction, gas recirculation rate, trash addition rate, and mode of operation. Important performance parameters to be measured are: minimization of undesired products – N2O, NH3, and volatile organic compounds; maximization of desired outcomes – BOD5 removal, CO2 production. A pre-existing aerobic composter will be adapted and engineering evaluation tests will be performed to ensure that the key design criteria were met. Two composter runs of baseline operation with a known solid waste feed will be conducted under aerobic vs. denitrification conditions using a given set of environmental and process conditions. We will then determine the effects of oxygen on composting of simulated space-mission trash (aerobic vs. denitrifying conditions vs. a combination of these). We will assess the need for an inoculum and develop both thermophilic and mesophilic denitrifying mixed cultures, if needed. These will be tested in denitrifying composter runs. The effects of nitrate addition rate on N2O production will be determined in sequentially replicated composter experiments as will the effects of mixing duration and frequency on nitrate distribution and on production of volatile fatty acids (VFAs). As many as 34 composter runs may be completed in the 18 month study period, if the duration of each run is 14 days (as per previous aerobic composter runs at KSC using a crop solid waste analog). To evaluate the feasibility of stabilizing trash from human space-habitation missions (STS, ISS, ALS) in a denitrifying composting system; to regenerate CO2 from biodegradable organic material; to stabilize trash and reduce health risks The project's overall goal is to determine if nitrate can be used as an electron acceptor to replace oxygen during composting of simulated space-mission food wastes to stabilize this solid waste.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of the research project is to determine the feasibility of using a denitrifying composter to stabilize trash from space-habitation (STS, ISS, ALS) life support activities, but the difficulty of supplying adequate levels of oxygen to compost municipal solid wastes (MSW) on Earth is also a problem. Instability of MSW is a result of easily biodegraded organic compounds, primarily from garbage / food wastes, including uneaten food. Biodegradation of these compounds under uncontrolled aerobic storage conditions leads to putrefaction with the concomitant production of foul-smelling odors and possibly to proliferation of potentially pathogenic microorganisms. Composting under denitrifying conditions could minimize these obnoxious odors as well as convert nitrogenous waste nitrate into innocuous nitrogen gas, thereby saving on treatment of Earth compost offgas products.
Task Progress:	Composter runs of baseline operation with a known solid waste feed were conducted under aerobic vs. denitrification conditions using a given set of environmental and process conditions. The effects of oxygen on composting of simulated space-mission trash (aerobic vs. denitrifying conditions vs. a combination of these) were determined. Inoculum was tested in denitrifying composter runs. The effects of nitrate addition rate on N2O production were determined in sequentially replicated composter experiments as will the effects of mixing duration and frequency on nitrate distribution and on production of volatile fatty acids (VFAs).
Bibliography Type:	Description: (Last Updated: 04/28/2011)
Abstracts for Journals and Proceedings	Strayer RF, Reid K, Rector TJ, Hummerick MP, Levine LH, Garland JL. "Feasibility of a Denitrification Composter to Stabilize Space-Mission Trash." Poster presentation at the 10th International Symposium on Microbial Ecology,Cancun, Mexico, Aug 22-27, 2004. 10th ISME Book of Abstracts, August 2004. p. 81. , Aug-2004
Papers from Meeting Proceedings	Strayer RF, Reid K, Rector TJ, Hummerick MP, Garland JL. "Design and Initial Tests of a Denitrification Composter to Stabilize Space-Mission Trash." 24th International Conference on Environmental Systems (ICES), Colorado Springs, CO, July 2004. 24th International Conference on Environmental Systems, Colorado Springs, CO, July 2004. SAE paper 2004-01-2465. http://dx.doi.org/10.4271/2004-01-2465 , Jul-2004